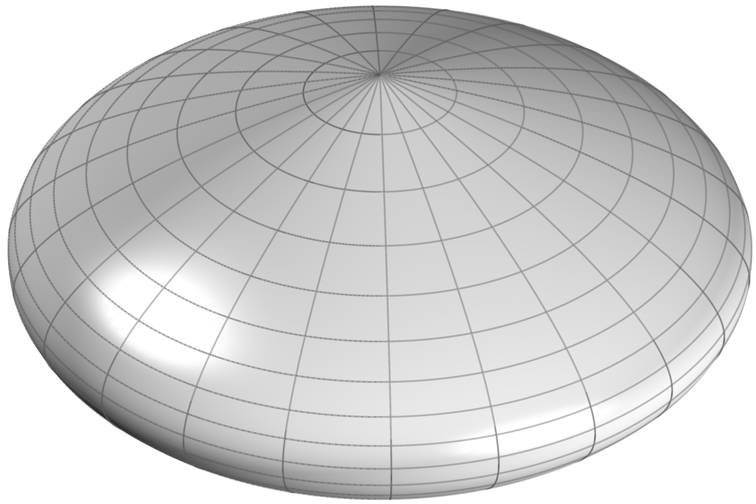

FIG. 1. Schematic representation of the complex parabolic metric manifold. Latitude lines correspond to spatial slices at fixed epoch, while longitude lines trace the temporal evolution of the manifold.

I. INTRODUCTION

Cosmological observations tightly constrain theoretical descriptions of the universe. Measurements of the cosmic microwave background (CMB) fix early-universe conditions and global geometry with high precision [1], while Type Ia supernovae (SNe Ia) probe the late-time expansion history through the luminosity distance–redshift relation [2]. Although each dataset is internally consistent, discrepancies arise when parameters inferred from early- and late-time observations are compared. In particular, the tension in the Hubble constant [1, 3] indicates that the standard cosmological model may be incomplete.

Historically, such discrepancies have been addressed by extending the model. The cosmological constant was introduced to permit a static universe [4] and abandoned following the discovery of expansion [5]. Galaxy rotation curves motivated the introduction of non-baryonic dark matter [6], and supernova evidence for accelerated expansion led to the reintroduction of the cosmological constant as dark energy [7, 8]. While these additions yield a phenomenologically successful framework, the dominant components of the cosmic energy budget remain physically unexplained [1]. More recent approaches introduce additional degrees of freedom, including time-varying dark energy [9], further increasing the parametric structure of the model.

We develop an alternative framework, termed Parabolic Metric Evolution (PME), in which large-scale evolution and local gravitational phenomena arise from a common geometric structure governed by a single intrinsic acceleration scale. The framework rests on two structural choices. First, separations are defined bilocally between ordered pairs of events rather than through a local infinitesimal metric, so that finite intervals remain well-defined even when the geometry evolves on the scale of the separation itself. Second, the manifold evolution is governed by a constant homogeneous acceleration A — a single intrinsic scale, the minimum required for nontrivial dynamics, whose value is determined empirically. The resulting homogeneous evolution yields a parabolic manifold extent, motivating the name Parabolic Metric Evolution.

These two choices fix the framework's predictions. The constant acceleration scale yields a circular-orbit envelope that reproduces the baryonic Tully–Fisher relation, with A determined empirically from galaxy rotation curves. Applied to the analytic distance–redshift relation that follows from the bilocal interval, the same scale predicts the Pantheon+ Type Ia supernova luminosity distances on the SH0ES-calibrated absolute scale, achieving a lower covariance-weighted χ^2 than Planck-calibrated FLRW. The present-day Hubble constant emerges as a prediction rather than a fitted parameter, and the acoustic angular scale at recombination is reproduced when the manifold kinematics are combined with standard recombination microphysics. No independent dark matter or dark energy sectors are required; the same acceleration scale governs galactic dynamics, cosmological expansion, and the baryonic density.

The framework differs from relativistic metric gravity in its bilocal definition of separation, and from modified-dynamics frameworks such as MOND [10, 11] in that the characteristic acceleration scale is not introduced phenomenologically but inherited from the global manifold kinematics. Cosmological evolution, gravity, and local dynamics are thereby unified through a single geometric scale.

II. BILOCAL GEOMETRY

A. Foundational Structure

The framework is defined by the following postulates:

1. Physical separations are defined bilocally between ordered pairs of events.

2. The manifold evolution is governed by a constant acceleration scale A.
3. The manifold is parameterized by an intrinsic evolution coordinate χ.
4. Observable time is defined operationally through null exchange, providing an invariant ordering of events. The intrinsic evolution parameter is related to this observable time by the projection $\chi = it$, which fixes the real–imaginary decomposition of temporal and spatial functionals and serves as a defining structural postulate of the framework.

Within this framework, geometric separation is defined operationally through ordered bilocal relations and possesses no independently specified continuum structure beneath this construction. All subsequent results follow from this structure.

B. Bilocal Spatial Separation

Let $p_e = (x_e^0, x_e^1)$ denote an emission event and $p_o = (x_o^0, x_o^1)$ an observation event, with the bilocal interval constructed from a temporal functional obtained from accumulated evolution between slices and a spatial functional defined from separations on the emission and observation slices.

The bilocal construction is illustrated in Figure 2. The spatial slices are closed, with $L(t)$ denoting the circumference of the slice at epoch t. The spatial slices scale with this global manifold extent, so the separations at emission and observation satisfy

$$\frac{\Delta x_e^1}{L(t_e)} = \frac{\Delta x_o^1}{L(t_o)}. \tag{1}$$

Imposing symmetry under interchange of emission and observation together with a smooth coincidence limit motivates adopting the midpoint average as the minimal symmetric choice,

$$\Delta s^1 = \Delta x_o^1 - \frac{1}{2}(\Delta x_o^1 - \Delta x_e^1), \tag{2}$$

$$\Delta s^1 = \frac{1}{2}(\Delta x_o^1 + \Delta x_e^1), \tag{3}$$

C. Invariance of the Bilocal Interval

Consider a bilocal scalar constructed from the temporal and spatial endpoint functionals $(\Delta s^0, \Delta s^1)$. We adopt a quadratic form as the lowest-order scalar consistent with finite

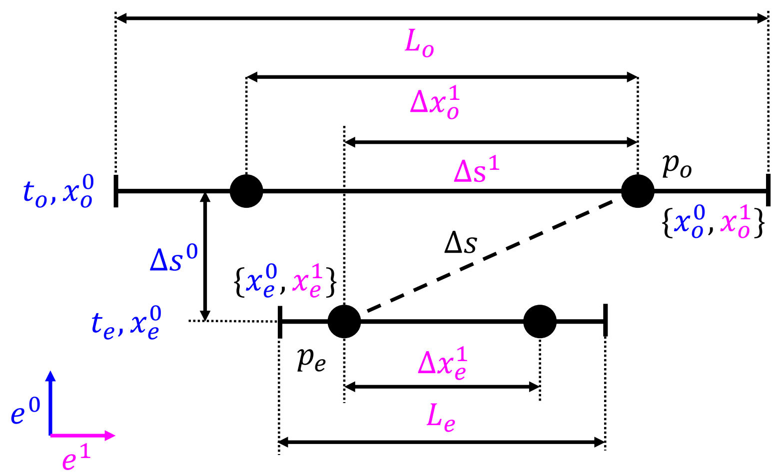

FIG. 2. Geometric construction of the bilocal interval between emission event p_e and observation event p_o, showing the accumulated temporal evolution and the symmetric midpoint spatial separation between scaled slices.

separations, ensuring that the local reduction yields a well-defined null condition while avoiding higher-order dependence on endpoint structure,

$$\Delta s^2 = a\,(\Delta s^0)^2 + b\,(\Delta s^1)^2 + 2c\,\Delta s^0 \Delta s^1. \tag{4}$$

Under interchange of the ordered endpoints (p_e, p_o), the temporal functional changes sign while the symmetric spatial functional does not, so the mixed term changes sign. Invariance therefore requires $c = 0$.

In the local reduction, sufficiently small endpoint separations render the temporal and spatial functionals linear in the coordinate increments. The null condition must define a finite null ratio, which requires both temporal and spatial contributions to enter non-degenerately so that the null condition defines a finite null ratio. The interval therefore reduces to

$$\Delta s^2 = a(\Delta s^0)^2 + b(\Delta s^1)^2.$$

The remaining quadratic form is therefore fixed up to an overall normalization and the relative weighting between the temporal and spatial functionals. These functionals are not arbitrary coordinates but are identified with the canonical imaginary and real components of the manifold, respectively. This identification fixes their relative normalization and selects the natural bilocal interval

$$\Delta s^2 = (\Delta s^0)^2 + (\Delta s^1)^2, \tag{5}$$

with Δs^0 purely imaginary and Δs^1 real.

D. Intrinsic Tangent Evolution

The intrinsic evolution of the manifold is parameterized by a coordinate χ. Observable time is introduced by the projection $\chi = it$, so that $d\chi = i\,dt$, fixing the real–imaginary decomposition used in the bilocal construction.

With constant intrinsic acceleration A, the tangent evolution scale along the manifold history follows from integration with respect to χ,

$$v_\chi(t) = \int A\,d\chi.$$

Substituting $d\chi = i\,dt$ gives $v_\chi(t) = i\int A\,dt$, which evaluates to

$$v_\chi(t) = i(At - V_0), \tag{6}$$

where V_0 is an integration constant representing the initial expansion rate of the manifold. The intrinsic tangent evolution is therefore purely imaginary, in accordance with the real–imaginary decomposition fixed by the projection $\chi = it$. Because the acceleration A acts homogeneously across the manifold, this tangent evolution is universal: every point on the manifold at epoch t shares the same intrinsic tangent generator $i(At - V_0)$. This tangent structure is physically real but nonobservable within the operational projection sector. Observable motion arises only through the real projection structure induced by the local null condition.

E. Induced Spatial Evolution

The observable spatial extent arises from integrating the intrinsic tangent evolution along the imaginary temporal direction. The global spatial scale of the manifold is therefore $L(t) \equiv \int v_\chi(t)\,d\chi$. Substituting the expression for $v_\chi(t)$ and using $d\chi = i\,dt$ gives $L(t) = \int i(At - V_0)\,i\,dt$. Evaluating the integral yields

$$L(t) = V_0 t - \frac{1}{2}At^2. \tag{7}$$

F. Manifold Deceleration

The intrinsic evolution is defined along χ, while observable dynamics are expressed in the projected time coordinate t with $d\chi = i\,dt$. Rewriting the manifold extent in terms of t yields its observable form. Differentiating twice with respect to t gives

$$\frac{d^2L}{dt^2} = -A, \tag{8}$$

so the manifold extent evolves with a constant kinematic deceleration of magnitude A, which is universal and independent of position or epoch.

III. COSMOLOGICAL DISTANCE

Observable cosmological distances follow from the bilocal interval defined in equation (5).

A. Temporal Separation

The temporal component of the bilocal interval is obtained from the accumulated tangent evolution along the manifold history between the emission and observation parameters t_e and t_o, $\Delta s^0 \equiv \int_{t_e}^{t_o} v_\chi(t)\,dt$. Substituting Eq. (6) gives

$$\Delta s^0 = \int_{t_e}^{t_o} i\left(At - V_0\right)\,dt. \tag{9}$$

Evaluating the integral yields

$$\Delta s^0 = i\left(\frac{A}{2}\left(t_o^2 - t_e^2\right) - V_0\left(t_o - t_e\right)\right). \tag{10}$$

The temporal separation is purely imaginary and depends only on the endpoints.

B. Spatial Separation

Solving equation (1) for the emission separation gives

$$\Delta x_e^1 = \Delta x_o^1\,\frac{L(t_e)}{L(t_o)}. \tag{11}$$

Because spatial slices scale with the manifold extent $L(t)$ and the slice separations satisfy the bilocal scaling relation given in equation (3), the spatial component becomes

$$\Delta s^1 = \frac{\Delta x_o^1}{2}\left(1 + \frac{V_0 t_e - \frac{1}{2}At_e^2}{V_0 t_o - \frac{1}{2}At_o^2}\right). \tag{12}$$

C. Bilocal Metric Formula

Substituting the temporal and spatial separations into equation (5) yields the interval relating the emission and observation events,

$$\Delta s^2 = \left[i \left(\frac{A}{2} \left(t_o^2 - t_e^2 \right) - V_0 \left(t_o - t_e \right) \right) \right]^2 + \left[\frac{\Delta x_o^1}{2} \left(1 + \frac{V_0 t_e - \frac{1}{2} A t_e^2}{V_0 t_o - \frac{1}{2} A t_o^2} \right) \right]^2 . \tag{13}$$

D. Redshift Relation

Redshift is defined observationally by the ratio of observed to emitted wavelength,

$$1 + z \equiv \frac{\lambda_o}{\lambda_e}. \tag{14}$$

We identify wavelength with a comoving spatial separation and assume that such separations scale with the manifold extent, so that $\lambda(t) \propto L(t)$.

$$1 + z = \frac{L(t_o)}{L(t_e)}. \tag{15}$$

Using equation (7), the emission epoch t_e is expressed in terms of t_o and the observed redshift z,

$$t_e = \frac{V_0(1+z) - \sqrt{(1+z)\left[(V_0 - A t_o)^2 + V_0^2 z\right]}}{A(1+z)}, \tag{16}$$

where the negative branch is chosen so that $t_e < t_o$.

Substituting into equation (13) and imposing the null condition $\Delta s^2 = 0$ yields the observable spatial separation,

$$D_C(z) = \Delta x_o^1 = \frac{t_o \left(2V_0 - A t_o\right) z}{2 + z}. \tag{17}$$

IV. CAUSAL STRUCTURE

The null structure of the bilocal interval also determines the extent of causal connectivity on the manifold. The largest spatial separation permitted by null ordering is obtained from equation (13) with the emission event at the origin of the manifold history, $t_e = 0$, and the observation event at epoch $t_o = t$. Solving the null condition $\Delta s^2 = 0$ for the observed separation gives the causal horizon

$$d_{\mathrm{ch}}(t) = t \left(2V_0 - At\right). \tag{18}$$

Differentiating the causal horizon gives the expansion rate of the causal boundary,

$$c_{\mathrm{c}}(t) = \frac{d}{dt}\, d_{\mathrm{ch}} = 2\,(V_0 - At)\,. \tag{19}$$

This quantity defines the rate at which the accessible causal domain can be extended under the operational clock, and therefore characterizes the growth of the causally orderable region.

Comparing the causal horizon with the manifold extent $L(t)$ defined in equation (7) yields

$$\frac{d_{\mathrm{ch}}(t)}{L(t)} = 2. \tag{20}$$

The causal horizon is therefore twice the manifold extent at all epochs, placing the entire last-scattering surface within a single causal region. Because the homogeneous manifold acceleration introduces no preferred direction, the large-scale isotropy of the cosmic microwave background follows directly from the geometry.

V. GRAVITY

When the temporal interval between neighboring events is small compared with the characteristic evolution scale of the manifold extent $L(t)$, the bilocal construction admits a smooth local reduction yielding an effective acceleration field inherited from the manifold kinematics.

A. Coincidence Limit

The local description is obtained as the coincidence limit of the bilocal interval as the endpoint separation tends to zero. Let the observation event occur at epoch t and the emission event at $t - \Delta t$, with $\Delta t \to 0$. To extract the leading nontrivial local structure, the temporal and spatial bilocal functionals are expanded to first and second orders in Δt.

Using the local expansion of the manifold extent,

$$L(t - \Delta t) = L(t) - \dot{L}(t)\,\Delta t - \frac{1}{2} A\,\Delta t^2,$$

the slice-separation relation gives

$$\frac{\Delta x_e^1}{L(t - \Delta t)} = \frac{\Delta x_o^1}{L(t)}. \tag{21}$$

Hence

$$\Delta x_e^1 = \Delta x_o^1 \left(1 - \frac{\dot{L}}{L}\Delta t\right), \tag{22}$$

so the bilocal spatial component becomes

$$\Delta s^1 = \Delta x_o^1 \left(1 - \frac{1}{2}\frac{\dot{L}}{L}\Delta t\right) \approx \Delta x_o^1 \tag{23}$$

to leading order in Δt. Over the same interval, the temporal component is

$$\Delta s^0 = \int_{t-\Delta t}^{t} i\,(At' - V_0)\;dt' \approx i\,(At - V_0)\,\Delta t = -i\,(V_0 - At)\,\Delta t. \tag{24}$$

Substituting these leading terms into equation (5) yields

$$\Delta s^2 \approx -\,(At - V_0)^2\,\Delta t^2 + (\Delta x_o^1)^2, \tag{25}$$

which is the emergent local interval obtained from the coincidence limit of the bilocal geometry. Although the intrinsic temporal contribution is imaginary, the squared interval is real, and the observable null scale arises from the operational projection condition that defines admissible signal exchange. Imposing $\Delta s^2 = 0$ on the coincidence-limit interval gives

$$(At - V_0)^2\,\Delta t^2 = \left(\Delta x_o^1\right)^2, \tag{26}$$

so the null ratio yields a real observable scale,

$$\frac{\Delta x_o^1}{\Delta t} = V_0 - At, \tag{27}$$

where the positive root is selected by the requirement that signal propagation proceed in the direction of increasing time on the expansion branch $At < V_0$; the contraction branch lies beyond the scope of the present treatment. The quantity $(V_0 - At)$ is therefore the observable null scale at epoch t, emerging as the real operational projection of the underlying imaginary tangent generator through the null condition that defines admissible exchange. The directed manifold structure itself is carried by the intrinsic tangent $i(At - V_0)$, whose integral curves define the temporal evolution structure of the manifold at each epoch.

B. Acceleration Field

Consider an event pair (p_e, p_o) whose temporal separation Δt satisfies $\Delta t \ll L/\dot{L}$. Expanding the manifold extent about a reference epoch t_0 gives

$$L(t_0 + \delta t) = L(t_0) + \dot{L}(t_0)\,\delta t + \frac{1}{2}\ddot{L}(t_0)\,\delta t^2 + \cdots\,. \tag{28}$$

Using equation (8), the quadratic term is governed by the constant local value $\ddot{L}(t_0) = -A$. The local reduction therefore contains a homogeneous second-order contribution corresponding to a uniform deceleration of magnitude A inherited from the global manifold evolution, which provides the physical origin of gravity in the local limit. This uniform deceleration defines the background acceleration field underlying local gravitational dynamics.

The effective inward acceleration in a bound system decomposes into a homogeneous background and a sourced concentration,

$$a_{\mathrm{in}}(x) = a_{\mathrm{bg}} + a_{\mathrm{conc}}(x). \tag{29}$$

The background term a_{bg} is uniform and has constant magnitude A. In the absence of localized content, it defines the manifold evolution, and an inertial frame follows it without inducing concentration or depletion. Localized inertial content, however, resists this acceleration, redistributing the flux into spatial concentrations that define the sourced gravitational component $a_{\mathrm{conc}}(x)$ governing bound dynamics. The inertial parameter m entering this redistribution is the same quantity that weights the invariant separation in the construction of geometric action, so that the gravitational response corresponds to a redistribution of the momentum-weighted action-space separation underlying the local geometry.

C. Minimal Local Closure

The bilocal construction fixes the homogeneous background acceleration scale A through the second-order local reduction of the manifold extent but does not determine the sourced concentration arising from spatial variations in inertial resistance. A weak-field closure is therefore required.

We adopt a minimal closure as the lowest-order local form consistent with locality, linearity in enclosed inertial content, rotational invariance, and additivity. These conditions restrict the sourced field to a divergence relation as the leading-order connection between field structure and inertial content, rather than an assumed inverse-square form. The concentration field therefore satisfies a divergence proportional to a local inertial density, with enclosed content given by its volume integral. In the weak-field limit, this density is dominated by rest mass and reduces to the baryonic density. All intrinsic kinematic scales are fixed by the acceleration scale A, while the constant G sets the coupling to localized inertial content without introducing an additional kinematic scale.

For a spherically symmetric configuration the sourced acceleration is radial,

$$\mathbf{a}_{\rm conc}(r) = a_r(r)\,\hat{r}. \tag{30}$$

In the local-reduction regime relevant for gravitationally bound systems, the concentration field resides in three spatial dimensions, so spherical symmetry is defined with respect to ordinary 2-spheres in $\mathbb{R}^3$. The flux through a sphere of radius r is

$$\Phi(r) = \oint_S \mathbf{a}_{\rm conc} \cdot d\mathbf{S} = 4\pi r^2 a_r(r). \tag{31}$$

Imposing flux conservation together with linearity in the enclosed inertial content gives

$$\oint_S \mathbf{a}_{\rm conc} \cdot d\mathbf{S} = -4\pi G M(r), \tag{32}$$

where $M(r)$ is the enclosed inertial content within radius r, obtained from the local density by

$$M(r) = \int_V \rho_{\rm inertial}\, dV, \tag{33}$$

and G is the empirical coupling relating inertial content to concentration of the background acceleration. Spherical symmetry then yields

$$\mathbf{a}_{\rm conc}(r) = -\frac{GM(r)}{r^2}\,\hat{r}, \tag{34}$$

and Gauss's theorem gives

$$\nabla\cdot \mathbf{a}_{\rm conc} = -4\pi G \rho_{\rm inertial}, \tag{35}$$

which is the Poisson equation for the sourced weak-field limit. In the weak-field regime relevant for galactic dynamics, the inertial density is dominated by baryonic rest-mass contributions, so $\rho_{\rm inertial} \approx \rho_b$.

D. Circular Orbits

For a body in a circular orbit of radius r with tangential speed v, the required centripetal acceleration is

$$a_{\rm cent}(r) = -\frac{v^2}{r}. \tag{36}$$

Equating this with the inward acceleration from equation (29) gives

$$-\frac{v^2}{r} = -A - \frac{GM_b(r)}{r^2}, \tag{37}$$

and multiplying by r gives

$$v^2 = Ar + \frac{GM_b(r)}{r}. \tag{38}$$

Then, solving for the enclosed baryonic mass gives

$$M_b(r) = \frac{r\,(v^2 - Ar)}{G}. \tag{39}$$

E. Circular-Orbit Solutions

Equation (39) defines the PME fundamental plane, a two-parameter family of circular-orbit solutions relating baryonic mass M, orbital radius r, and orbital speed v.

For fixed orbital speed v, the mass function $M(r)$ is a concave quadratic in r. Differentiating gives

$$\frac{dM}{dr} = \frac{v^2 - 2Ar}{G}. \tag{40}$$

The mass therefore reaches a maximum at

$$r_{\max} = \frac{v^2}{2A}. \tag{41}$$

Substituting this radius into equation (39) yields the maximum baryonic mass compatible with a circular orbit at speed v,

$$M_{\max}(v) = \frac{v^4}{4AG}. \tag{42}$$

The surface $M(v, r)$ therefore defines a fundamental plane of circular-orbit solutions, as shown in Figure 3, with the ridge $M_{\max}(v)$ giving the upper envelope of baryonic mass for a given orbital speed. Galaxies dominated by circular rotation are expected to saturate this boundary and follow a characteristic mass–velocity relation.

F. Baryonic Tully–Fisher Relation

$M_{\max}(v)$ defines the scaling $M \propto v^4$, consistent with the observed baryonic Tully–Fisher relation [12].

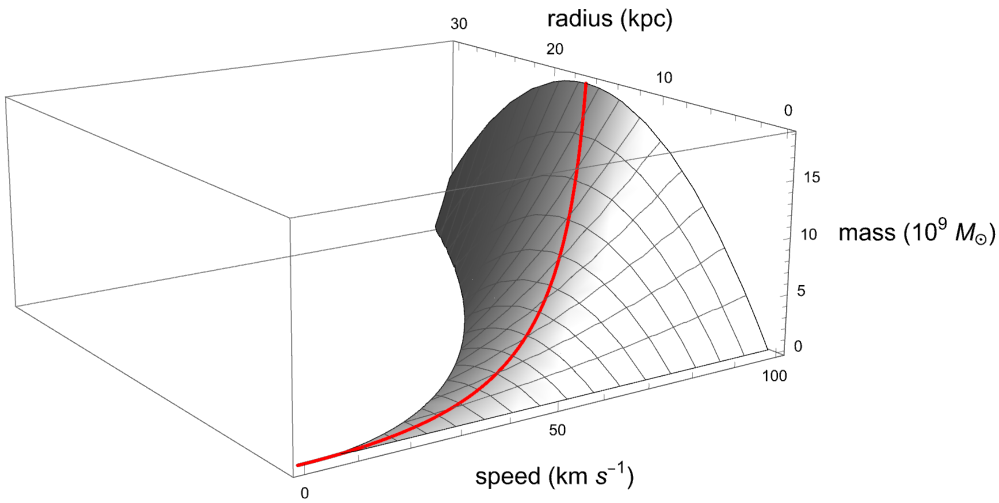

FIG. 3. PME fundamental plane defined by equation (39). The baryonic mass surface $M(v, r)$ is shown as a function of orbital speed v and orbital radius r for an illustrative manifold acceleration scale $A = 10^{-11}\,\mathrm{m\,s^{-2}}$. The red ridge marks the locus $M_{\mathrm{max}}(v)$ corresponding to the maximum baryonic mass permitted by the manifold kinematics at each orbital speed. Galaxies populating this ridge reproduce the baryonic Tully–Fisher relation.

A characteristic acceleration scale in galaxy dynamics has previously been emphasized in frameworks such as MOND [10], in which the force law is modified to reproduce Tully–Fisher-type scaling.

The manifold acceleration A is determined empirically by fitting the circular-orbit envelope (equation (42)) to the SPARC galaxy sample of McGaugh, Lelli and Schombert [11], using the assumptions $\Upsilon_\star = 0.5$, $f_{\mathrm{gas}} = 1.33$, zero intrinsic mass-to-light uncertainty, and the quality cut Qual ≤ 3. These rotationally supported disk systems are well suited to the analysis, as their kinematics closely follow the circular-orbit approximation.

For each galaxy the rotation curve provides an asymptotic circular velocity v, while the baryonic mass M_b is computed as $M_b = \Upsilon_\star L_{3.6} + f_{gas}\, M_{\mathrm{HI}}$. The value of A is determined by minimizing the χ^2 statistic between the theoretical envelope and the observed (M_b, v) pairs. The best-fit manifold acceleration is

$$A = 3.7 \times 10^{-11}\,\mathrm{m\,s^{-2}}. \tag{43}$$

The observed galaxies follow the predicted $M \propto v^4$ scaling across a wide range of baryonic masses and rotational velocities, as shown in Figure 4. The baryonic Tully–Fisher rela-

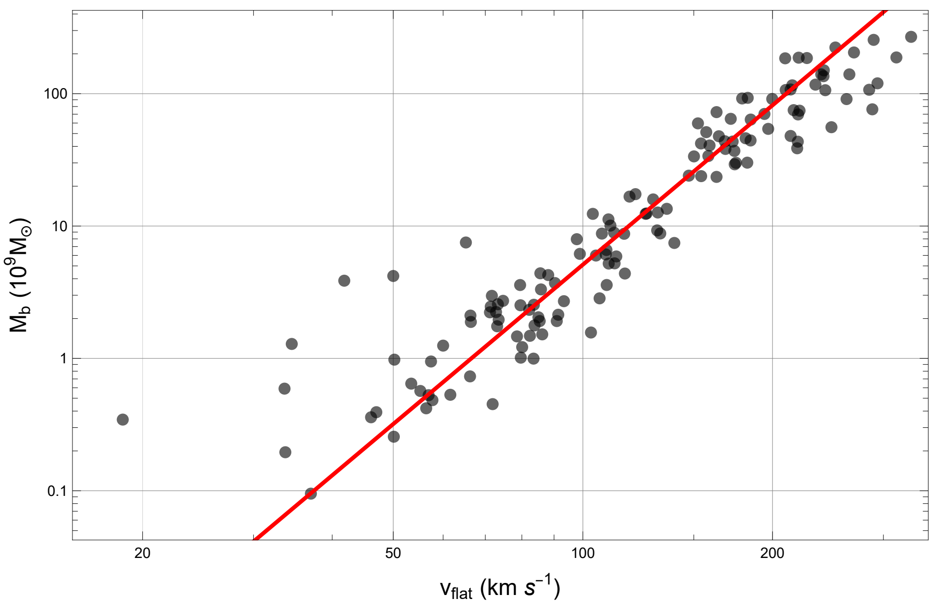

FIG. 4. Log–log plot of baryonic mass M_b versus outer rotation speed v for the SPARC galaxy sample of McGaugh, Lelli and Schombert [11]. Black circles show the observed galaxies. The solid red line shows the PME circular-orbit envelope evaluated at the best-fit acceleration.

tion is a direct consequence of the constant-acceleration structure of the manifold and an observational signature of the acceleration flux.

G. Global Acceleration Budget

The local closure fixes all non-homogeneous acceleration flux as sourced by inertial content. In the weak-field regime this reduces to baryonic mass. Integrating the flux law over the full homogeneous domain of scale L gives

$$\Phi_{\rm tot} = \oint \mathbf{a}_{\rm conc} \cdot d\mathbf{S} = -4\pi G M_b^{\rm tot}. \tag{44}$$

Using the global relation between baryonic mass and the background acceleration scale,

$$M_b^{\rm tot} = \frac{AL^2}{G}, \tag{45}$$

yields

$$\Phi_{\rm tot} = -4\pi A L^2. \tag{46}$$

Thus the available concentration flux is fixed by the background scale A and is completely accounted for by baryonic mass. Local gravitational systems therefore correspond to partial redistributions of a finite acceleration budget, so gravitational collapse approaches a saturated configuration rather than an unbounded one.

H. Mach's Bucket

The bilocal reduction introduces a homogeneous acceleration scale A fixed by the geometry of the manifold. This acceleration is not sourced by baryonic matter; instead, baryonic mass redistributes it through conservation of acceleration flux (equation (32)).

For a rotating system of radius r and tangential speed v, circular motion requires an inward centripetal acceleration v^2/r. In this framework, that requirement is met by redistributing the background field to produce the necessary inward concentration.

Because the total acceleration is finite, this redistribution cannot be local: any concentration within the system must be balanced globally. Rotational inertia therefore depends on the global distribution of mass through the shared acceleration field, providing a Machian interpretation.

I. Regularity of the Manifold

The finite acceleration budget established by equation (32) fixes the total flux available for redistribution by inertial content. Gravitational concentration is therefore the partial redistribution of a finite reservoir, and collapse approaches saturated configurations of finite density and radius rather than singular limits.

Causal ordering is intrinsic to the bilocal structure: separations are defined between ordered event pairs, and the operational clock counts null round-trips that accumulate monotonically along the manifold tangent. Trajectories that would close back on themselves are excluded by this monotonicity, so the framework admits no closed timelike curves and no associated causal paradoxes.

VI. LOCAL DYNAMICS

Having established the gravitational and causal structure, we now define the operational dynamics governing motion within the manifold. The treatment here establishes the kinematic foundation; further development of time-dilation phenomena and their geometric origin in the acceleration flux structure is outside the scope of this article.

A. Operational Time

Imposing the null condition $\Delta s^2 = 0$ on equation (25) gives

$$(V_0 - At)^2 \Delta t^2 = \Delta x^2, \tag{47}$$

so that

$$\Delta t = \frac{\Delta x}{(V_0 - At)}. \tag{48}$$

This defines the null boundary of the local geometry and fixes the relation between spatial separation and temporal increment through the observable null scale $(V_0 - At)$. The null condition therefore defines the observable projection of the underlying imaginary manifold tangent through the observable null scale $(V_0 - At)$.

Observable time is defined operationally by null exchange between nearby worldlines. One tick of the clock is a complete round trip of a signal along null trajectories, and the temporal parameter t is identified with the number of such cycles, which defines an invariant ordering of events. The null relation ensures that each cycle corresponds to a temporal increment determined by the local scale $(V_0 - At)$ at that epoch, and restricts admissible trajectories to those consistent with this ordering.

The null relation therefore fixes the local causal structure and operational ordering of events. Observable motion and the invariant timelike interval both arise as real operational projections of this same local geometry, so causality, motion, and measurable duration are unified through the null structure of the manifold.

B. Proper Time

Using the local interval defined in equation (25),

$$ds^2 = -(V_0 - At)^2\, dt^2 + dx^2, \tag{49}$$

proper time defines the invariant timelike interval along admissible trajectories, $d\tau^2 \equiv -ds^2$, so that

$$d\tau^2 = (V_0 - At)^2\, dt^2 - dx^2, \tag{50}$$

giving

$$d\tau = (V_0 - At)\sqrt{1 - \frac{v^2}{(V_0 - At)^2}}\, dt, \tag{51}$$

where $v = dx/dt$. Proper time therefore represents the timelike operational projection of the underlying imaginary tangent generator onto admissible observable trajectories.

C. Local Momentum

The homogeneous manifold acceleration A selects a timelike direction, and the local interval (equation (25)) decomposes the imaginary tangent evolution into orthogonal temporal and spatial components. Observable local geometry therefore resolves as real operational projections of this common tangent structure, with orientation parameterized by an angle θ relative to the flux.

Let $d\lambda$ denote an affine increment along the imaginary tangent, an invariant manifold parameter from which observable projections derive. Resolving the local projection structure induced by the tangent yields

$$dx = (V_0 - At)\sin\theta\, d\lambda, \qquad dt = \cos\theta\, d\lambda, \tag{52}$$

with the observable null scale $(V_0 - At)$ setting the projection scale per unit increment of λ at epoch t. Measurable velocity is then the ratio of the spatial and temporal projections,

$$v = \frac{dx}{dt} = (V_0 - At)\tan\theta, \tag{53}$$

and observable momentum follows by inertial weighting,

$$P = m\, v = P_{\text{geom}} \tan\theta, \tag{54}$$

where $P_{\text{geom}} \equiv m(V_0 - At)$ is the geometric momentum scale associated with the observable null projection of the manifold tangent at epoch t.

The null boundary is reached when $v = V_0 - At$, requiring $\tan\theta = 1$ and therefore $\theta = \pi/4$. At this orientation the spatial and temporal projections are equal in magnitude, and observable motion is restricted to the timelike domain $0 \le \theta < \pi/4$. Beyond this boundary the projection geometry remains well-defined through θ, but the kinematic description in terms of t ceases to apply; its continuation is defined in terms of phase evolution.

D. Local Action

The bilocal construction admits a natural system of units in which mass and energy are reciprocally dimensioned along the manifold tangent. Mass is rendered in units of time per unit length, while energy comes in units of length per unit time. Under this convention, the product $m(V_0 - At)$ is dimensionless and energy has units of velocity, same as the manifold tangent. The dimensional conversion constants between inertial and recurrence quantities are therefore absorbed into the geometry rather than introduced separately.

In the coincidence limit, the local interval (equation (25)) defines the observable projection structure induced by the underlying imaginary tangent evolution together with an invariant interval ds describing local manifold separation. Weighting this invariant separation by the geometric momentum scale

$$P_{\text{geom}} \equiv m(V_0 - At), \tag{55}$$

defines the invariant geometric action increment,

$$dS_{\text{geom}} \equiv P_{\text{geom}}\, ds, \tag{56}$$

with quadratic form

$$dS^2_{\text{geom}} = P^2_{\text{geom}}\, ds^2. \tag{57}$$

Using the local invariant

$$ds^2 = -(V_0 - At)^2 dt^2 + dx^2, \tag{58}$$

the action-space interval becomes

$$dS^2_{\text{geom}} = m^2 (V_0 - At)^2 \left[-(V_0 - At)^2 dt^2 + dx^2\right]. \tag{59}$$

The temporal component defines the geometric energy scale associated with the observable projection of tangent evolution.

$$E_{\text{geom}}(t) \equiv -m(V_0 - At)^2, \tag{60}$$

while the spatial component is governed by the same geometric momentum scale $P_{\text{geom}} = m(V_0 - At)$.

Evaluating the local interval along the projected tangent trajectory gives $ds^2 = -(V_0 - At)^2 d\lambda^2$, which is timelike on the expansion branch. Taking the positive root yields the invariant element along the tangent,

$$ds = (V_0 - At)\, d\lambda, \tag{61}$$

and therefore

$$dS_{\rm geom} = m(V_0 - At)^2\, d\lambda, \tag{62}$$

so the invariant geometric action accumulates along the projected manifold trajectory,

$$S_{\rm geom} = \int dS_{\rm geom}. \tag{63}$$

Under the geometric unit convention established above, $dS_{\rm geom}$ inherits the dimensions of invariant separation itself. Action is therefore expressed geometrically in units of length, so any minimum action scale corresponds geometrically to a minimum manifold separation rather than an independently imposed dimensional constant.

The invariant geometric action $dS_{\rm geom}$ admits a projection into the observable timelike sector, defining an observable action increment dS. Resolving the tangent evolution into its temporal and spatial contributions expresses this projection in canonical form,

$$dS = P\, dx - H\, dt, \tag{64}$$

which identifies P as the observable momentum conjugate to x and H as the observable Hamiltonian conjugate to t. The Hamiltonian H coincides with the geometric energy scale $E_{\rm geom}$ defined in equation (60). The observable action dS therefore admits a Hamilton–Jacobi interpretation within the timelike projection sector, while $dS_{\rm geom}$ remains the invariant quantity defined across the full manifold.

Since $dS_{\rm geom}$ defines the momentum-weighted invariant separation of the manifold, physical trajectories follow the shortest paths in action space. The principle of least action therefore emerges as the geometric shortest-distance condition on the invariant action geometry.

E. Recurrence Energy

The operational clock counts ordered null recurrences between neighboring worldlines, with one tick corresponding to a complete round trip along null trajectories. For an event pair separated by Δx, the null relation gives

$$\Delta t = \frac{\Delta x}{(V_0 - At)}, \tag{65}$$

so the corresponding recurrence rate is

$$\nu = \frac{1}{\Delta t} = \frac{V_0 - At}{\Delta x}. \tag{66}$$

Multiplying the recurrence rate by the associated separation recovers the invariant tangent scale,

$$\Delta x\, \nu = V_0 - At. \tag{67}$$

Accumulated duration and local recurrence density therefore arise from the same underlying exchange structure, differing only in whether recurrence is integrated or measured locally.

Because geometric separation is defined only through ordered bilocal relations, the framework does not admit infinitesimal separations as fundamental geometric objects. The local continuum instead arises as a coincidence-limit approximation of finite bilocal relations rather than as an independently defined substructure beneath them. Operational recurrence therefore possesses physical meaning only across a minimum bilocal separation, denoted h. Evaluating the recurrence structure at this minimum scale defines the recurrence energy,

$$E_{\text{rec}} \equiv h\nu. \tag{68}$$

The recurrence energy is therefore the recurrence rate weighted by the minimum bilocal separation defining the operational geometry. At the operational floor defined by h, the recurrence relation becomes $\nu = (V_0 - At)/h$, so the energy relation reduces to the identity

$$E_{\text{rec}} = h\nu = V_0 - At, \tag{69}$$

so that at the operational resolution limit the recurrence-weighted energy coincides with the observable null projection induced by the underlying manifold tangent generator. The minimum separation h is therefore not an independently introduced constant but the scale at which the recurrence structure of the bilocal geometry reduces directly to the underlying tangent invariant.

Under the geometric unit convention established above, both the recurrence scale h and the invariant action S_{geom} inherit the dimensions of geometric separation itself. Planck recurrence and action therefore arise as geometric length scales within the operational projection structure.

The geometric energy scale $E_{\text{geom}}(t) \equiv -m(V_0 - At)^2$ established in the action geometry is the inertially weighted form of the same invariant tangent structure. Using equation (69), the two forms are related by the geometric momentum scale $P_{\text{geom}} = m(V_0 - At)$,

$$E_{\text{geom}} = -P_{\text{geom}}\, E_{\text{rec}}, \tag{70}$$

so the recurrence and inertial forms of energy arise from the same invariant tangent structure, differing only by the inertial weighting of the projected trajectory. The Planck–Einstein relation emerges as the operational recurrence relation at the bilocal resolution scale, while the inertial form arises from the momentum-weighted action geometry.

F. Phase Evolution

The projection-based kinematic description applies within the timelike sector, where the local invariant defines a real proper-time interval. At the null boundary, $ds^2 = 0$ and $d\tau \to 0$, marking the limit of observable timelike motion. Beyond this boundary, observable motion ceases, while the invariant geometric structure remains well-defined and the geometric action continues to accumulate along the manifold tangent.

Beyond the null boundary, the underlying imaginary tangent generator persists even though observable timelike parametrization ceases within the real operational sector. Phase therefore parametrizes the continued accumulation of invariant geometric action beyond the proper-time sector. Because the operational clock is defined through ordered null recurrence, one complete operational bounce subtends 2π radians of accumulated phase. The corresponding action-per-radian conversion is therefore

$$\hbar \equiv \frac{h}{2\pi}, \tag{71}$$

where h is the minimum operationally meaningful bilocal separation defined by the operational floor identity (69). Phase evolution is then given by

$$\phi = \frac{S_{\text{geom}}}{\hbar}. \tag{72}$$

Combined with equation (69), this gives $\hbar\omega = V_0 - At$ at the operational floor, with $\omega = 2\pi\nu$, recovering the angular form of the Planck–Einstein relation as a geometric closure condition on the operational cycle rather than as a postulated quantization. The distinction between

h and $\hbar$ is physical rather than notational in the present framework: h defines the minimum operational separation underlying the recurrence structure of the manifold, while $\hbar$ converts accumulated geometric action into radian-valued phase through closure of the operational cycle.

Phase evolution therefore represents the continuation of invariant manifold evolution beyond the proper-time-parametrized sector, while the underlying geometric structure remains continuous across the null boundary.

VII. LATE-TIME EVOLUTION

The distance–redshift relation derived from the bilocal interval predicts the late-time expansion history.

A. Type Ia Supernova Sample

We use the Pantheon+ compilation of 1701 spectroscopically confirmed Type Ia supernovae presented by Brout et al. [2]. The publicly released Pantheon+SH0ES dataset provides the observed distance moduli μ_{obs}, redshifts z, and the full STAT+SYS covariance matrix for covariance-weighted χ^2 inference. The distance moduli are absolutely calibrated through the SH0ES determination of the Type Ia absolute magnitude M_B.

B. Luminosity Distance

The spatial separation predicted by the manifold geometry is given by equation (17). The luminosity distance follows from the standard relation $D_L(z) = (1+z)D_C(z)$. Substituting the spatial separation yields the corresponding analytic luminosity distance

$$D_L(z) = -\frac{t_o\,(At_o - 2V_0)\,z(1+z)}{2+z}. \tag{73}$$

C. Distance Modulus

Type Ia supernova observations are reported in terms of the distance modulus, which relates the observed luminosity distance to the measured brightness of each supernova,

$$\mu(z) = 5\log_{10}\left(\frac{D_L(z)}{10\,\mathrm{pc}}\right). \tag{74}$$

D. Manifold Kinematic Parameters

The local null scale at the observation epoch is

$$c_o = V_0 - At_o. \tag{75}$$

Operationally, photon exchange is used as a proxy for this local null scale, so the measured speed of light at the observation epoch determines c_o, fixing the integration constant,

$$V_0 = c_o + At_o. \tag{76}$$

With the manifold acceleration A determined from the baryonic Tully–Fisher analysis, the Cepheid-calibrated Pantheon+ subset is used to determine the observation epoch t_o on the absolute distance scale. For each candidate value of t_o, the corresponding V_0 follows algebraically from equation (76), and theoretical distance moduli $\mu_{\mathrm{th}}(z)$ are evaluated at the calibration-set redshifts. These are compared with the observed values $\mu_{\mathrm{obs}}(z)$, defining the residual vector

$$r_i = \mu_{\mathrm{obs}}(z_i) - \mu_{\mathrm{th}}(z_i). \tag{77}$$

The best-fit observation epoch is obtained by minimizing the covariance-weighted chi-square statistic

$$\chi^2 = r^{\mathrm{T}} C^{-1} r, \tag{78}$$

where C is the full Pantheon+ STAT+SYS covariance matrix. The resulting best-fit value of t_o then determines V_0, completing the kinematic parameter set. The resulting parameters are listed in Table I.

E. Comparison with Observations

Figure 5 shows the PME prediction and the Planck 2018 FLRW model compared with the Pantheon+SH0ES data.

Parameter	Value
A	$3.7 \times 10^{-11}\,\mathrm{m\,s^{-2}}$
V_0	$3.16 \times 10^{8}\,\mathrm{m\,s^{-1}}$
t_o	$4.51 \times 10^{17}\,\mathrm{s}$

TABLE I. PME kinematic parameters determined from BTFR and Cepheid-calibrated Pantheon+ subset.

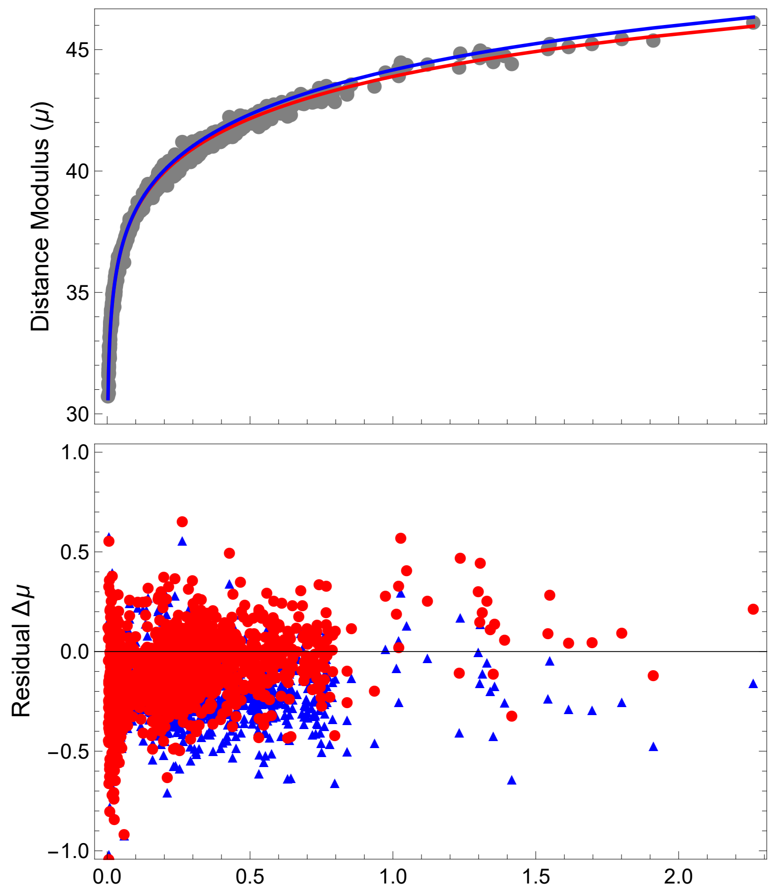

FIG. 5. Pantheon+SH0ES distance moduli as a function of redshift z, compared with the PME prediction and the spatially flat FLRW model. PME parameters are fixed by BTFR and Cepheid calibration. FLRW parameters are fixed by the Planck 2018 results.

The Pantheon+SH0ES dataset provides observed distance moduli $\mu_{\mathrm{obs}}(z)$ calibrated on an absolute scale through the SH0ES determination of the Type Ia supernova absolute magnitude. This calibration is applied identically to both models.

Each model predicts $D_L(z)$, which is mapped to $\mu_{\rm th}(z)$ using the definition of $\mu(z)$ introduced earlier. We compare $\mu_{\rm th}(z)$ to $\mu_{\rm obs}(z)$ using the full Pantheon+ STAT+SYS covariance matrix.

PME is evaluated using the fixed parameter set derived above. FLRW is evaluated using a spatially flat model specified by the Planck 2018 parameter set $\Omega_m = 0.315$, $\Omega_\Lambda = 1-\Omega_m$, and $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [1]. The comparison is performed on the out-of-sample supernovae, excluding the Cepheid calibration records.

The resulting covariance-weighted chi-square values are

$$\chi^2_{\rm PME} = 2726, \qquad \chi^2_{\rm FLRW} = 4049, \tag{79}$$

for $\nu = 1625$, corresponding to reduced values $\chi^2/\nu = 1.68$ and 2.49, respectively. PME predicts the observed distance moduli substantially more accurately than the Planck 2018 FLRW model across the Pantheon+SH0ES sample, without dark sector parameters.

F. Hubble Function

The Hubble expansion rate is defined by the fractional rate of change of the real spatial extent of the manifold, $H(t) \equiv \dot{L}(t)/L(t)$. Differentiating equation (7) gives $\dot{L}(t) = V_0 - At$, so

$$H(t) = \frac{V_0 - At}{V_0 t - \frac{1}{2}At^2}. \tag{80}$$

Evaluating equation (80) at the observation epoch $t = t_o$ using the best-fit manifold parameters gives the present-day expansion rate

$$H_0 = 66.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{81}$$

The Hubble constant therefore emerges as a prediction rather than a fitted parameter.

G. Cosmic Time

The redshift–time relation differs from standard cosmological evolution due to the underlying parabolic evolution. Table II compares the resulting cosmic times with those from FLRW algorithms.

The parabolic expansion yields uniformly larger cosmic ages than FLRW, significantly extending the available time for early structure formation.

z	PME [Myr]	FLRW [Myr]
0	14,283	13,791
1	7,039	5,840
10	1,265	470
100	137	16.4
1,000	13.9	0.429

TABLE II. Cosmic time as a function of redshift. PME predictions are compared with FLRW values computed using Planck 2018 parameters [1].

VIII. EARLY-TIME EVOLUTION

Independent constraints arise from the cosmic microwave background (CMB), whose acoustic scale probes the distance to last scattering relative to the sound horizon.

The expansion history is determined by the PME kinematics derived above, while recombination microphysics is evaluated using standard rate equations. Thermodynamic quantities are mapped using the scale factor

$$a(t) = \frac{L(t)}{L(t_o)}. \tag{82}$$

A. Baryon and Photon Densities

The recombination history and pre-decoupling sound speed require the baryon and photon densities. The baryon density is fixed by the manifold acceleration scale rather than introduced as an independent cosmological parameter. This follows from the local divergence law (equation (35)), which relates the sourced acceleration field to the baryonic density.

For a homogeneous baryonic distribution of density ρ_b within a region of characteristic size L, the enclosed mass is

$$M(L) = \frac{4\pi}{3} L^3 \rho_b, \tag{83}$$

so the corresponding baryon-sourced acceleration at radius L is

$$\frac{GM(L)}{L^2} = \frac{4\pi}{3} G L \rho_b. \tag{84}$$

Evaluating this expression at the present manifold extent $L(t_o)$, which contains the full homogeneous baryonic source, and identifying it with the inferred manifold acceleration A

gives

$$A = \frac{4\pi}{3} G L(t_o) \rho_b, \tag{85}$$

hence

$$\rho_b = \frac{3A}{4\pi G\, t_o \left(V_0 - \frac{1}{2} A t_o\right)}. \tag{86}$$

Using the kinematic parameters determined in the previous sections yields

$$\rho_b = 9.49 \times 10^{-28}\,\mathrm{kg\,m^{-3}}. \tag{87}$$

This value is consistent with independent observational estimates of the present-day baryon density from primordial light-element abundances [13].

The photon density is determined from the measured CMB temperature $T_\gamma = 2.725\,\mathrm{K}$. The mapping between temperature and photon density depends on the local geometric scale $(V_0 - At)$, so the photon density is treated as epoch dependent rather than constant. For a blackbody radiation field, the photon number density is

$$n_\gamma(t) = \frac{2\zeta(3)}{\pi^2} \left(\frac{k_B T_\gamma}{\hbar (V_0 - At)} \right)^3, \tag{88}$$

and, following the geometric sign convention of Eq. (60), the mean photon energy is

$$\langle E_\gamma \rangle = -\frac{\pi^4}{30\zeta(3)} k_B T_\gamma. \tag{89}$$

The corresponding photon energy density is

$$u_\gamma(t) = n_\gamma(t) \langle E_\gamma \rangle, \tag{90}$$

so the mass-equivalent photon density becomes

$$\rho_\gamma(t) = \frac{u_\gamma(t)}{-(V_0 - At)^2}. \tag{91}$$

These densities determine the baryon–photon momentum ratio and the free-electron density entering the visibility function.

B. Recombination Epoch

The recombination epoch is the time of maximum photon visibility, i.e. the most probable last-scattering time in the baryon–photon plasma.

To determine the ionization history, we evaluate the standard recombination rate equations using `Recfast++` with the PME expansion geometry. The background evolution is provided by the PME Hubble function (equation (80)) in place of the FLRW algorithm, with the expansion rate governed by the best-fit manifold parameters. The baryon sector is specified by the present-day baryon density derived above. The resulting ionization history is used to interpolate the free-electron fraction $x_e(z)$ entering the visibility function. The free-electron number density $n_e(t)$ is defined as

$$n_e(t) = x_e(t)\,\frac{\rho_b}{m_p}\,a(t)^{-3}, \tag{92}$$

where ρ_b is the present-day baryon mass density, m_p is the proton mass.

The Thomson scattering rate is expressed in terms of the local particle density, cross section, and local geometric scale,

$$\Gamma_T(t) = n_e(t)\,\sigma_T\,(V_0 - At), \tag{93}$$

where σ_T is the Thomson cross section. The optical depth is then defined as the accumulated interaction probability along the manifold history,

$$\tau_T(t) = \int_t^{t_o} \Gamma_T(t')\,dt'. \tag{94}$$

The corresponding visibility function is

$$g(t) = \Gamma_T(t)\,e^{-\tau_T(t)}. \tag{95}$$

Evaluating $g(t)$ on the PME expansion background using the `Recfast++` microphysics parametrization yields a maximum at $z_* = 1039$, $t_* = 13.4\,\mathrm{Myr}$ shown in Fig. 6.

C. Sound Horizon

Prior to recombination, baryons and photons form a tightly coupled fluid whose acoustic interaction distance defines the sound horizon. The thermodynamic evolution follows the scale factor $a(t)$ defined in equation (82).

The baryon–photon momentum ratio is

$$R(t) = \frac{3\rho_b}{4\rho_\gamma(t)}\,\frac{a(t)^{-3}}{a(t)^{-4}} = \frac{3\rho_b}{4\rho_\gamma(t)}\,a(t), \tag{96}$$

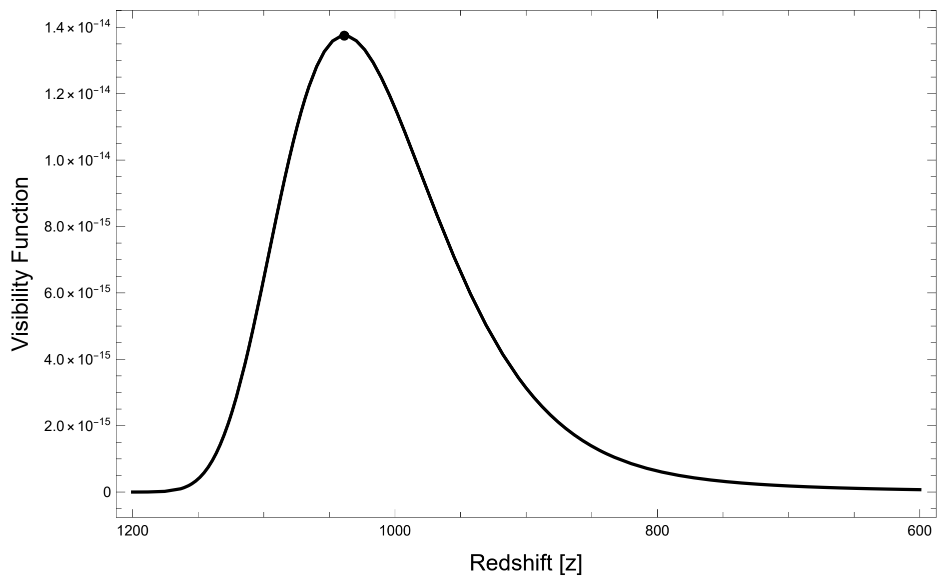

FIG. 6. Photon visibility function $g(z)$ evaluated using the recombination history on the PME expansion background. The peak defines the recombination epoch, yielding $z_* = 1039$.

giving the sound speed

$$c_s(t) = \frac{c_{\rm c}(t)}{\sqrt{3\,(1 + R(t))}}. \tag{97}$$

The causal expansion rate $c_{\rm c}(t)$ sets the rate at which interactions can be ordered through the operational clock and thus defines the effective causal scale entering the acoustic horizon integral, so that the sound horizon is

$$r_s = \int_0^{t_*} c_s(t)\, dt. \tag{98}$$

Evaluating this expression using the manifold parameters yields $r_s = 3.7\,\mathrm{Mpc}$.

D. Acoustic Angular Scale

The observed acoustic angle is determined by the ratio of the sound horizon at recombination to the bilocally assigned transverse curvature scale of the emission–observation pair. Because the spatial slices are closed with circumference $L(t)$, the acoustic feature at recombination spans the full transverse extent of the last-scattering slice, and the observer integrates over the full transverse extent of the present slice. The bilocal pair relevant to this observation is therefore the maximally-bracketed configuration in which the transverse endpoints on each slice reach the full slice circumference, so that $\Delta x^1_e = L(t_*)$ and $\Delta x^1_o = L(t_o)$.

Applying the bilocal spatial rule of equation (3) to this maximally-bracketed transverse configuration defines the bilocal-averaged transverse circumference,

$$C_\perp \equiv \frac{1}{2}\left[L(t_o) + L(t_*)\right]. \tag{99}$$

Using the redshift relation from equation (15), this becomes

$$C_\perp = \frac{L(t_o)}{2}\,\frac{2+z_*}{1+z_*}. \tag{100}$$

Angular features on a closed transverse surface subtend angles relative to its radius of curvature. For the bilocal pair the effective transverse radius is therefore obtained from the bilocal-averaged circumference,

$$r_\perp = \frac{C_\perp}{2\pi} = \frac{L(t_o)}{4\pi}\,\frac{2+z_*}{1+z_*}. \tag{101}$$

The acoustic angular scale is then

$$\theta_* = \frac{r_s}{r_\perp}. \tag{102}$$

Substituting the expression above for $r_\perp$ gives

$$\theta_* = \frac{4\pi(1+z_*)}{L(t_o)(2+z_*)}\,r_s. \tag{103}$$

Using equation (7) evaluated at t_o yields

$$\theta_* = \frac{8\pi(1+z_*)}{t_o(2V_0 - At_o)(2+z_*)}\,r_s. \tag{104}$$

Evaluating this expression with the manifold parameters gives

$$\theta_* = 0.0103. \tag{105}$$

Expressed in the conventional CMB notation,

$$100\,\theta_* = 1.03. \tag{106}$$

This result follows from the PME expansion geometry combined with standard recombination microphysics.

IX. DISCUSSION

The minimum bilocal separation h introduced in the construction of geometric energy is not an independently imposed discreteness postulate, but follows from the operational definition of the geometry itself. Because geometry is defined only through ordered bilocal relations, the framework contains no independently defined sub-operational continuum in which infinitesimal separations possess autonomous geometric meaning. The local continuum appears only as the coincidence-limit approximation of fundamentally finite bilocal relations. The divergence of the recurrence density in the coincidence limit therefore reflects the breakdown of the operational description as the bilocal separation ceases to remain physically meaningful. The bilocal geometry admits a minimum operational separation, represented by h, at which recurrence and tangent structure coincide.

This same local structure provides the basis for the dynamical interpretation of the framework. Observable time, proper time, momentum, action, energy, and phase are not introduced as independent physical primitives, but arise as projections or continuations of the invariant tangent geometry inherited from the bilocal construction. In particular, the momentum-weighted invariant separation defines an action geometry in which physical trajectories follow shortest paths in action space. The principle of least action is therefore recovered as a geometric shortest-distance condition rather than imposed as an independent variational postulate.

A detailed nucleosynthesis calculation has not been attempted within the present framework. Standard Big Bang nucleosynthesis treatments assume time-independent microphysics, including fixed particle rest energies and equilibrium thermodynamic mappings. In PME, however, the effective manifold energy scale varies explicitly with cosmic time through the tangent structure, and the operational reaction clock need not coincide with the geometric expansion time. These features break the time-translation invariance implicit in conventional BBN calculations, so the reaction network would need to be re-derived within the PME framework, which is outside the scope of this article.

This distinction also affects baryon asymmetry. In equilibrium with time-reversal symmetric, time-independent microphysics, matter–antimatter annihilation drives the net baryon number toward zero, leaving a radiation-dominated universe. Standard cosmology therefore treats the baryon-to-photon ratio as an external parameter rather than deriving it from Standard Model physics. The explicitly time-dependent evolution of the PME energy scale relaxes the equilibrium assumptions underlying this argument, permitting a nonzero residual baryon density without invoking additional symmetry-breaking mechanisms. PME therefore provides a framework in which baryon asymmetry may arise from the underlying geometric evolution, allowing nucleosynthesis to be formulated self-consistently within this context. A complete formulation is outside the scope of this article.

Extending the bilocal framework to perturbations, structure growth, nucleosynthesis, and a fully intrinsic treatment of recombination is required to assess viability beyond the homogeneous background solution but is outside the scope of this article. The present formulation establishes a constrained kinematic structure whose physical adequacy must ultimately be judged by its ability to reproduce observational phenomena across these additional domains.

The framework predicts a residual non-Keplerian contribution to bound motion with radial scaling Ar^2/GM. In compact systems this effect is suppressed and becomes significant only at larger radii. Solar-System observations therefore constrain the model but do not probe the regime in which the background term becomes dynamically dominant. Improved precision, particularly at large heliocentric distances, should reveal systematic departures from purely Keplerian motion. Detection or exclusion of such a signal provides a direct test of the construction.

A. Code Availability

A computational notebook implementing the bilocal geometry, distance–redshift relation, and observational analysis is publicly available at:

`https://www.wolframcloud.com/obj/32b2e831-2cbe-4ff5-b821-aa23f476f015`

The notebook reproduces the results and figures presented in this work and is provided to enable independent verification of the calculations.

X. CONCLUSION

A constrained bilocal construction on a complex manifold yields a quadratic interval whose local reduction produces a homogeneous acceleration scale. This single geometric scale governs both the parabolic evolution of the manifold and the redistribution of acceleration flux in gravitationally bound systems. The bilocal geometry produces an emergent local null constraint, defines an invariant action geometry, and admits a real operational projection structure from which observable time, proper time, momentum, energy, and phase arise.

In the local-reduction regime, action is not an independently postulated functional but the momentum-weighted invariant separation of the manifold. Physical trajectories therefore follow shortest paths in action space, so the principle of least action emerges as a geometric consequence of the bilocal construction. Phase evolution provides the continuation of invariant tangent evolution beyond the proper-time-parametrized sector, while the operational recurrence structure identifies the role of h and $\hbar$ within the geometry rather than introducing them as independent dimensional inputs.

The evolution implied by this construction yields a closed analytic distance–redshift relation, from which the Pantheon+ Type Ia supernova luminosity distances follow directly on an absolute scale. The same parameter set predicts the present-day Hubble constant and produces uniformly extended cosmic ages. In the local-reduction regime, redistribution of the homogeneous acceleration field generates an inverse-square sourced component and a circular-orbit envelope consistent with the observed baryonic Tully–Fisher relation. The acceleration scale inferred from galaxy dynamics then fixes the baryonic density of the homogeneous universe, linking galactic and cosmological observables without introducing non-baryonic dark matter or dark energy components.

The bilocal construction also determines the causal horizon and implies a constant ratio between the causal domain and manifold extent, placing the last-scattering surface within a single causal region and accounting for large-scale isotropy. The geometry yields a transverse scale for recombination that reproduces the observed acoustic angular scale. Because the available acceleration flux is finite, gravitational collapse approaches saturated configurations, eliminating curvature singularities and enforcing a globally ordered causal structure. Rotational inertia emerges from redistribution of the same finite acceleration budget, providing a Machian interpretation in which local inertial behavior depends on the global distribution of inertial content.

Evolution, causal structure, action, gravity, inertia, recurrence, and phase therefore arise from a common bilocal origin, with the framework's local kinematic and gravitational structure arising from the momentum-weighted invariant separation. The framework provides a unified explanation of isotropy, galaxy rotation, Type Ia supernova magnitudes, the acoustic scale, nonsingular gravitational collapse, the baryonic density, and the Hubble constant, while remaining directly testable through galactic dynamics, high-redshift distance measurements, gravitational lensing, Solar-System residuals, and the requirement that a complete early-universe treatment reproduce observed nucleosynthesis.

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